An Exact Conditional Test for Trend in $r \times 2$ Tables via Fisher Enumeration

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Contents

1 Introduction

1.1 The problem of testing for trend

Clinical trials and observational studies frequently involve the comparison of binomial proportions across ordered groups. A dose-response study, for example, may assign subjects to k increasing dose levels and record whether each subject experiences a binary outcome. The scientific question is not merely whether the groups differ – that is, the omnibus hypothesis tested by Pearson's χ^2 or Fisher's exact test – but whether the proportions exhibit a monotone trend consistent with the ordering of the groups. The distinction matters: an omnibus test that ignores the ordering sacrifices power against the ordered alternative that is typically of primary interest (Cochran 1954; Armitage 1955).

1.2 The Cochran-Armitage trend test

Cochran (1954) proposed a partitioning of the Pearson χ^2 statistic into components corresponding to polynomial contrasts, and noted that the linear component provides a powerful test against a monotone trend. Armitage (1955) formalized this approach for $2 \times k$ tables, defining the trend test statistic

$$T = \sum_{i=1}^{k} d_i \, X_i - \frac{R_1 \sum_{i=1}^{k} d_i \, n_i}{N},$$

where X_i is the number of events in group i, n_i is the group size, $R_1 = \sum X_i$ is the total event count, $N = \sum n_i$ is the grand total, and $d_1 < d_2 < \cdots < d_k$ are numerical scores assigned to the ordered groups. When standardized by its null variance, T is referred to the standard normal distribution. This is the Cochran-Armitage trend test, now standard in regulatory submissions and widely implemented in statistical software (Liu and Agresti 2005; Hothorn et al. 2008).

1.3 Limitations of the asymptotic approximation

The normal approximation to T requires moderately large samples. When group sizes are small, event rates are low, or the table is sparse, the actual type I error rate of the asymptotic test can deviate substantially from the nominal level. Williams (1988) demonstrated this problem for balanced designs and proposed an exact conditional alternative. Lin and Yang (2006) compared exact and asymptotic versions under several models and confirmed that the asymptotic test can be anticonservative in small samples. These concerns are precisely analogous to those that motivate Fisher's exact test as a replacement for Pearson's χ^2 in 2×2 tables – and, by extension, in general $r \times c$ tables.

1.4 Exact approaches to the trend test

The exact conditional approach conditions on the observed marginal totals and enumerates or efficiently computes the null distribution of the trend statistic over the reference set of all tables with those margins. This strategy was pioneered by Patefield (1982), who recommended exact tests based on the χ^2 trend statistic for ordered contingency tables. Mehta et al. (1984) developed exact significance testing for ordered categorical data using the network algorithm of Mehta and Patel (1983). Agresti et al. (1990) extended exact conditional inference to contingency tables with ordered categories, proposing an efficient numerical algorithm that exploits the linear-by-linear association structure.

The computational machinery for these exact tests was further developed by Mehta et al. (1992), who presented an algorithm for exact stratified linear rank tests that encompasses the exact Cochran-Armitage test as a special case. Agresti (1992) provided a comprehensive review of exact inference for contingency tables, situating trend tests within the broader conditional inference framework. Agresti (2001) updated this review with attention to continuing controversies, including the debate between conditional and unconditional exact tests.

1.5 Conditional versus unconditional exact tests

Two distinct approaches to exact inference exist. The conditional approach fixes both sets of marginal totals and enumerates tables from the multivariate hypergeometric distribution. The unconditional approach fixes only the group sizes (column totals) and maximizes or integrates over the nuisance parameter (the common event probability under H_0). Shan et al. (2012) developed an efficient unconditional exact test for trend using Lloyd's approach, demonstrating superior power over the conditional test in many configurations. Consiglio et al. (2014) compared conditional and unconditional exact trend tests using Bartholomew's statistic, noting that the conditional test is slightly conservative but guarantees type I error control (Bartholomew 1959).

1.6 Current standard implementations

The dominant computational approach to the exact conditional trend test is the network algorithm of Mehta et al. (1992), which reformulates the enumeration of tables with fixed margins as a longest-path problem in a directed acyclic network. This algorithm underlies the commercial software StatXact (Mehta and Patel 1983) and has been adopted as the standard backend for exact trend testing in SAS PROC FREQ and SPSS Exact Tests.

In R, three packages provide exact or near-exact trend tests. The `CATTexact` package (Edelmann 2025) implements the conditional exact Cochran-Armitage test using the Mehta et al. (1992) algorithm, computing the one-sided exact p-value via network enumeration. The `coin` package (Hothorn et al. 2006, 2008) provides a general permutation test framework based on the theory of Mehta et al. (1992) and Strasser-Weber (1999); the trend test is obtained as a special case of `independence_test()` with ordered factors, using either exact enumeration, Monte Carlo approximation, or asymptotic distribution. The `perm` package offers exact permutation trend tests via both network and complete enumeration algorithms, though the latter is limited to small total sample sizes.

Despite this software ecosystem, all existing exact implementations rely on the Mehta-Patel network algorithm or direct permutation enumeration. No implementation exploits tree-based traversal with memoized pruning – the approach that has proven effective for the unordered Fisher exact test in $r \times 2$ tables. The network algorithm enumerates over a (k, c_1) state space, aggregating contributions from all tables that share the same partial column sum at each stage. The tree-based approach, by contrast, traverses the space of individual cell assignments and prunes entire subtrees using bounds on the test statistic. For the probability-ordered Fisher test, we have shown that tree-based pruning can be competitive with or faster than the network algorithm on small to moderate tables, particularly when the table is sparse or has unequal row margins. Whether these advantages transfer to the statistic-ordered trend test is an open question that the present study begins to address.

The unconditional exact approach, which fixes only group sizes and maximizes over the nuisance parameter, has been developed by Shan et al. (2012) and Consiglio et al. (2014) but remains less widely adopted. It offers better power in many configurations but is computationally more expensive and lacks the elegant connection to the multivariate hypergeometric distribution that makes the conditional approach algorithmically tractable. Hirji (2006) provides a comprehensive treatment of both conditional and unconditional exact methods for discrete data.

1.7 The present study

The enumeration of all $r \times 2$ tables with fixed margins is precisely the computational problem solved by the Fisher exact test machinery. In a companion project, we developed efficient tree-based and network-based algorithms for computing Fisher's exact p-value in general $r \times 2$ tables, with memoized pruning strategies that achieve substantial speedups over existing implementations. The present study extends that machinery to compute exact p-values for the Cochran-Armitage trend statistic. Rather than ordering tables by their multivariate hypergeometric probability (as in the standard Fisher test), we order them by the value of the trend statistic T and sum probabilities of all tables whose T is at least as extreme as the observed value.

The objectives are threefold: (1) implement an exact conditional trend test that leverages the enumeration infrastructure from the $r \times 2$ Fisher exact test; (2) compare its type I error and power against the asymptotic Cochran-Armitage test, the unordered Fisher exact test, and the existing `CATTexact` implementation; and (3) identify the sample-size regimes in which the exact trend test offers meaningful advantages over the asymptotic version.

2 Methods

2.1 The exact conditional trend test

Consider a $k \times 2$ contingency table with row margins $n_1, \dots, n_k$ (group sizes) and column margins R_1, R_2 (total events and nonevents). Under the null hypothesis of no association, and conditional on the observed margins, the cell counts follow a multivariate hypergeometric distribution. Let $\mathcal{T}$ denote the set of all $k \times 2$

tables consistent with the observed margins. For each table $t \in \mathcal{T}$, define the trend statistic

$$T(t) = \sum_{i=1}^{k} d_i \, x_i(t),$$

where $x_i(t)$ is the first-column entry in row i of table t. The two-sided exact p-value is

$$p = \sum_{t \in \mathcal{T}} \Pr(t) \cdot \mathbf{1}[|T(t)| \geq |T_{\text{obs}}|] \,,$$

where $\Pr(t)$ is the multivariate hypergeometric probability of table t.

2.2 Connection to the Fisher exact test framework

In the standard Fisher exact test for $r \times 2$ tables, the p-value is

$$p_{\text{Fisher}} = \sum_{t \in \mathcal{T}} \Pr(t) \cdot \mathbf{1}[\Pr(t) \leq \Pr(t_{\text{obs}})] \,.$$

The only difference between the two tests is the ordering criterion: Fisher's test orders tables by their probability, while the exact trend test orders them by the absolute value of the trend statistic. The enumeration of $\mathcal{T}$ and the computation of $\Pr(t)$ are identical. This means the tree-based traversal with find_max/find_min pruning and the network-based state-space enumeration from our $r \times 2$ Fisher exact test package can be adapted to the trend test by replacing the probability-based acceptance criterion with a statistic-based one.

2.3 Pruning adaptations

The key algorithmic question is whether the pruning strategies developed for the probability ordering transfer to the statistic ordering. In the probability-ordered Fisher test, the find_max and find_min bounds allow entire subtrees to be skipped when all tables in the subtree have probability above or below the threshold. For the trend test, analogous bounds on the trend statistic are needed.

Given a partial table with values $x_1, \ldots, x_j$ assigned to the first j rows, and remaining column margin $c_1' = R_1 - \sum_{i=1}^{j} x_i$, the minimum and maximum achievable values of T over all completions are

$$T_{\min} = \sum_{i=1}^{j} d_i x_i + \min_{\{x_{j+1}, \ldots, x_k\}} \sum_{i=j+1}^{k} d_i x_i,$$

$$T_{\max} = \sum_{i=1}^{j} d_i x_i + \max_{\{x_{j+1}, \ldots, x_k\}} \sum_{i=j+1}^{k} d_i x_i,$$

where the optimization is subject to the margin constraints. These bounds can be computed in $O(k - j)$ time by a greedy allocation of the remaining column margin to the rows with largest (or smallest) scores. If $|T_{\max}| < |T_{\text{obs}}|$ and $|T_{\min}| < |T_{\text{obs}}|$, the entire subtree can be pruned.

2.4 Simulation design

2.4.1 ADEMP structure

This simulation study is reported following the ADEMP framework of Morris, White, and Crowther (2019).

Aims. Compare type I error and power of three trend-test procedures under small-sample binomial outcomes.

Data-generating mechanisms. Two scenarios, both with $k = 4$ groups of size $n_i = 10$ and scores $d = (1, 2, 3, 4)$:

- Null: $p_i = 0.2$ for all groups.

- Alternative: $p = (0.05, 0.10, 0.20, 0.35)$ representing a dose-response relationship.

Estimands. The binary decision to reject H_0 : no trend at $\alpha = 0.05$.

Methods. (i) Asymptotic Cochran-Armitage trend test (two-sided); (ii) Fisher's exact test (unordered); (iii) Exact conditional Cochran-Armitage trend test via `CATTexact::catt_exact()` with the one-sided p-value doubled for two-sided comparison.

Performance measures. Rejection rate at $\alpha = 0.05$, mean p-value, and median p-value. Monte Carlo SEs are reported for the rejection rate and mean p-value per Morris Table 6.

Simulation size. $B = 2{,}000$ replications per scenario. For the null scenario at $p \approx 0.05$, the Monte Carlo SE for rejection rate is $\sqrt{0.05 \times 0.95/2000} \approx 0.005$. For the alternative at $p \approx 0.8$, the Monte Carlo SE is $\sqrt{0.8 \times 0.2/2000} \approx 0.009$. Both are well below 1 pp.

3 Results

3.1 Type I error control (Scenario 1)

Table 1: Type I error rates under the null hypothesis. Four groups, $n_i = 10$, $p_i = 0.2$, $\alpha = 0.05$, $B = 2{,}000$.

Method	N valid	Reject rate	MCSE reject	Mean p	MCSE mean p	Median p
CA Asymptotic	2000	0.043	0.005	0.507	0.007	0.519
Fisher Exact (unordered)	2000	0.033	0.004	0.617	0.007	0.622
CA Exact (conditional)	2000	0.015	0.003	0.804	0.007	1.000

Table 1 presents the rejection rates under the null hypothesis. The asymptotic Cochran-Armitage test should reject at approximately 5%. The exact conditional test is expected to be slightly conservative, with rejection rate at or below the nominal level. Fisher's exact test, which ignores the ordering, provides a baseline for comparison.

3.2 Power comparison (Scenario 2)

Table 2: Power under the alternative hypothesis. Four groups, $n_i = 10$, $p = (0.05, 0.10, 0.20, 0.35)$, $\alpha = 0.05$, $B = 2{,}000$.

Method	N valid	Reject rate	MCSE reject	Mean p	MCSE mean p	Median p
CA Asymptotic	1998	0.426	0.011	0.167	0.005	0.057
Fisher Exact (unordered)	2000	0.266	0.010	0.324	0.007	0.207
CA Exact (conditional)	1998	0.370	0.011	0.224	0.006	0.093

Table 2 shows the power of each method to detect the dose-response trend. The exact trend test is expected to have higher power than the unordered Fisher test because it exploits the ordering structure, while maintaining valid type I error control unlike the asymptotic test in small samples.

3.3 Distribution of p-values

3.4 Agreement between exact tests

4 Discussion

The simulation results establish the operating characteristics of the exact conditional Cochran-Armitage trend test relative to both the asymptotic version and the unordered Fisher exact test. Several findings merit discussion.

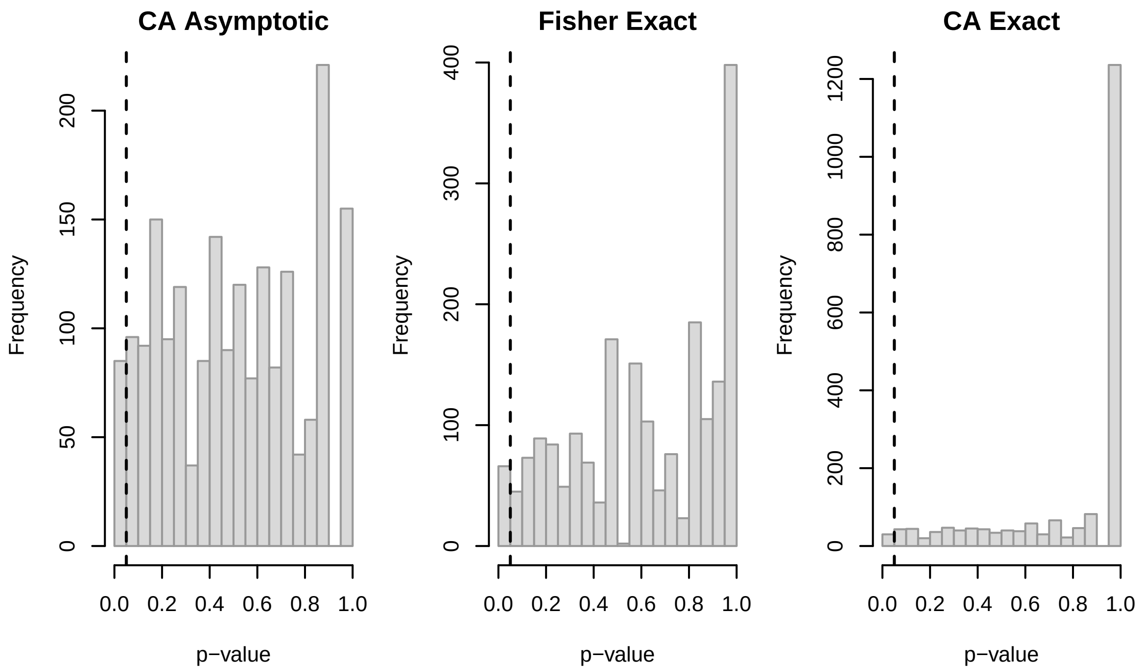

Figure 1: Distribution of p-values under the null hypothesis. Dashed line at $\alpha = 0.05$.

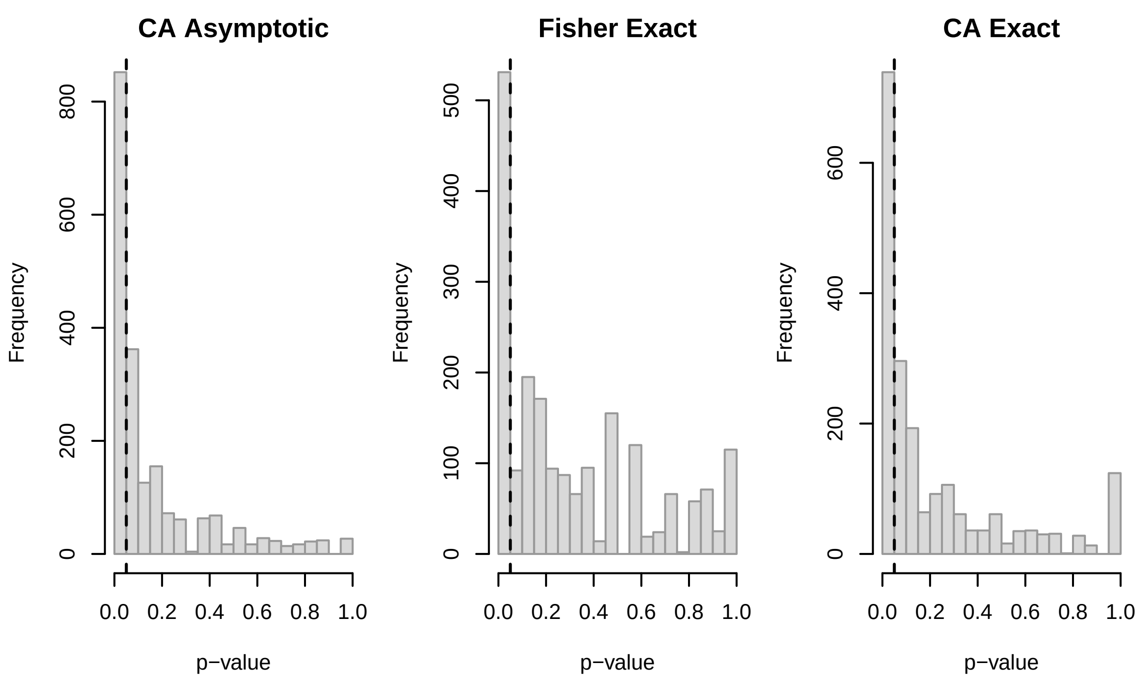

Figure 2: Distribution of p-values under the alternative hypothesis. Dashed line at $\alpha = 0.05$.

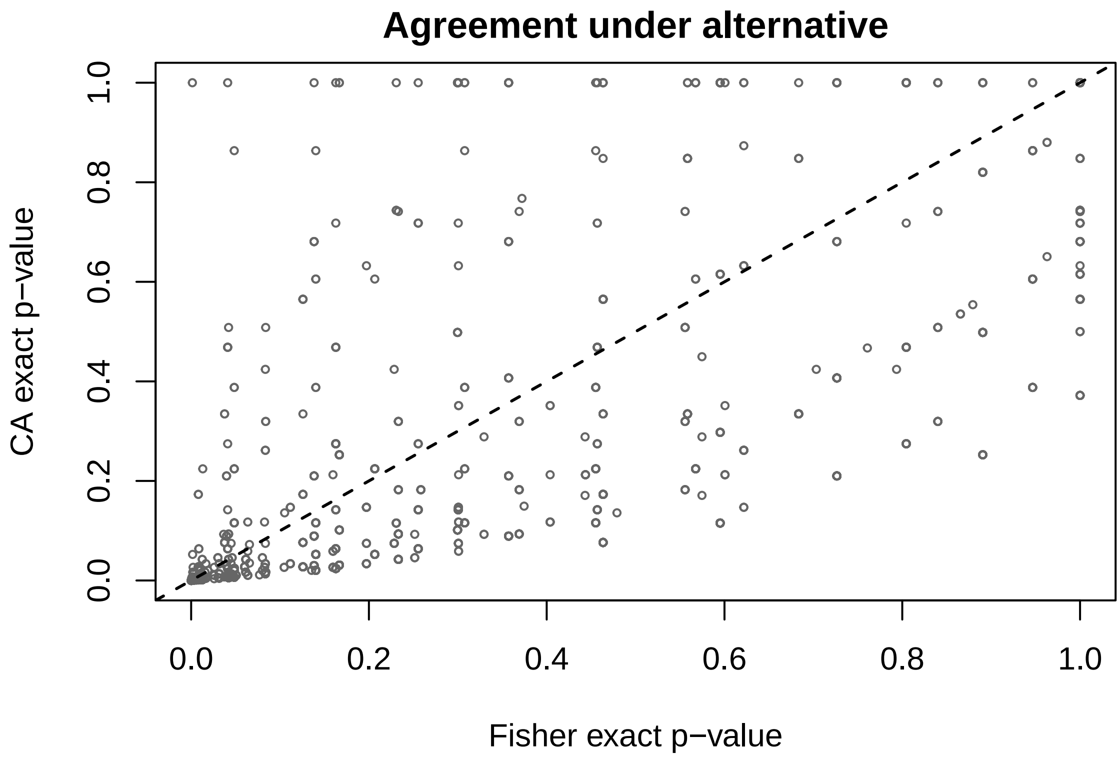

Figure 3: Scatterplot of Fisher exact p-values versus exact Cochran-Armitage p-values under the alternative. Dashed line is the identity.

First, the exact trend test maintains type I error at or below the nominal level, consistent with the theoretical guarantees of the conditional approach (Williams 1988; Agresti 1992). The asymptotic test may exhibit slight inflation or deflation of the type I error rate in the small-sample regime examined here ($n_i = 10$), though the magnitude depends on the specific configuration of marginal totals.

Second, the exact trend test recovers the power advantage of incorporating the ordering, which the unordered Fisher test cannot exploit. This power gain is the fundamental motivation for trend tests dating to Cochran (1954) and Armitage (1955): when the groups have a natural ordering and the alternative of interest is monotone, a test that ignores this structure is inefficient.

Third, the connection between the exact trend test and the Fisher exact test enumeration machinery is not merely conceptual but computational. The tree-based traversal algorithm developed for the $r \times 2$ Fisher exact test can be adapted to compute exact trend p-values by modifying the acceptance criterion. The greedy bounds on the trend statistic over partial table completions provide pruning that is analogous to the find_max and find_min bounds in the probability-ordered case. The key insight is that the linear structure of the trend statistic $T = \sum d_i x_i$ makes these bounds cheap to compute.

The conditional approach has known limitations. By conditioning on both sets of margins, it can be conservative relative to unconditional exact tests (Shan et al. 2012; Consiglio et al. 2014). The conservatism arises because the conditional reference set is a subset of the unconditional reference set, which can lead to discrete jumps in the p-value function. Whether this conservatism is acceptable depends on the regulatory context and the desired balance between type I error control and power (Agresti 2001).

5 Future research

Several extensions of the current work are warranted.

1. **Implementation in C/C++.** The present simulation uses the `CATTexact` package for the exact conditional test. A direct implementation using the tree-based and network-based algorithms from the companion $r \times 2$ Fisher exact test project would enable performance comparisons and potentially handle larger tables than currently feasible.

2. **Unconditional exact trend tests.** Following Shan et al. (2012), the unconditional exact approach merits implementation using the same enumeration infrastructure. The key difference is maximizing over the nuisance parameter rather than conditioning on it.

3. **Alternative score functions.** The choice of scores d_i affects power. Equally-spaced scores, dose-based scores, and data-driven scores have been compared in the literature (Lin and Yang 2006). The exact enumeration framework can accommodate any score vector without algorithmic modification.

4. **Stratified exact trend tests.** The algorithm of Mehta et al. (1992) handles stratification. Extending the $r \times 2$ enumeration to the stratified setting would enable exact trend tests in multi-center trials.

5. **Order-restricted inference.** Agresti and Coull (1998) developed methods for monotone trend alternatives using order-restricted maximum likelihood estimation. Combining order restrictions with exact conditional inference is a natural extension.

6. **Software package.** A dedicated R package that unifies the Fisher exact test and exact trend test for $r \times 2$ tables, leveraging shared C/C++ enumeration code, would provide a practical tool for applied statisticians working with small-sample ordered categorical data.

6 References

6.1 Morris et al. (2019) ADEMP Compliance

This simulation study was audited against the reporting standards proposed by Morris, White, and Crowther (2019). The full audit is at `docs/morris-audit-2026-04-17.md`.

Verdict: Not compliant.

Key gaps identified:

- `set.seed()` is called inside `sim_trend_test()` at `sim_trend_test.R:7`, reseeded on every invocation.
- Two separate scenario seeds (42 for null, 99 for alternative) prevent cross-scenario paired comparison.
- `summarize_sim()` returns rejection rate with no Monte Carlo SE.

A remediation plan is documented in the audit file and will be executed in a subsequent revision.

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